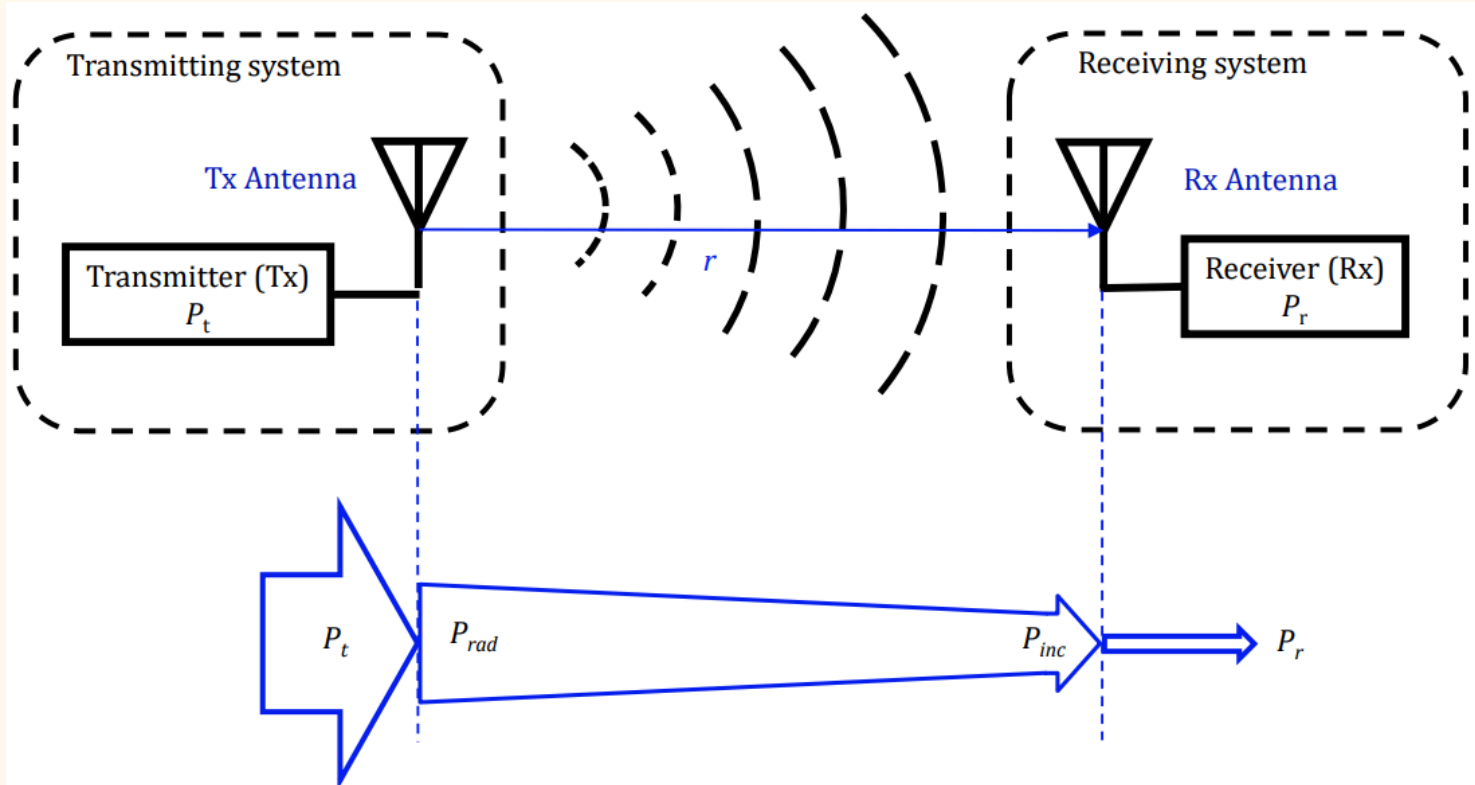


RF Link



Matching Efficiency (only for TL)

$$e_m = \frac{P_A}{P_t} = 1 - |\Gamma|^2 \quad (\text{only for TL})$$

best match is when $R_A = Z_0$

For lumped from power source

$$P_A = P_S \frac{R_A}{R_A + R_S}$$

best match is when $R_A = R_S^*$

Antenna Efficiency (Realised Radiation Efficiency)

$$e_{ant} = \frac{P_{rad}}{P_t} = e_m e_{rad}$$

Power

$$P_{rad} = e_{ant} P_t$$

$$P_{reflected} = |\Gamma|^2 P_t$$

P_t = transmitter output power

* exclude internal power loss, so no need 1/2 for perfect conjugate match

P_A = antenna input power

$$P_{A/loss/rad} = \frac{1}{2} |I|^2 R_{A/loss/rad}$$

Gain (Absolute/IEEE Gain)

$$G(\theta, \phi) = e_{rad} D(\theta, \phi)$$

Realised Gain

$$G_{realised}(\theta, \phi) = e_{ant} D(\theta, \phi)$$

EIRP

$$ERIP = G_{realised, t}(\theta, \phi) P_t$$

Pathloss

$$PL = \frac{P_{rx}}{P_{tx}} = G_{tx} G_{rx} \left(\frac{\lambda}{4\pi r}\right)^2 e_{pm}$$

* Gains are realised

* $PL < 1$

Freespace Pathloss

$$FSPL = \left(\frac{4\pi r}{\lambda}\right)^2$$

$$FSPL_{dB} = 20 \log\left(\frac{4\pi r}{\lambda}\right)$$

* $FSPL > 1$

Effective Aperture/Area, in m^2

$$A_e(\theta, \phi) = \frac{P_{rx}}{U_{incident}}$$

$$A_e(\theta, \phi) = G_{realised} A_{e, iso}$$

$$A_e(\theta, \phi) = G_{realised}(\theta, \phi) \frac{\lambda^2}{4\pi}$$

if given a table of A and G for multiple antenna at the same frequency, plot graph to find gradient and discard outlier

Aperture Efficiency

$$e_{ap} = \frac{A_{effective}}{A_{physical}} = \frac{G_{realised}}{G_{max}}$$

Polarisation Matching Efficiency

$$e_{pm} = |\hat{a}_{incident} \cdot \hat{a}_{rx}|^2 = \cos^2 \theta$$

$\hat{a}$ = direction of E field

Received Power

$$P_{rx} = U_{incident} A_{e, rx} e_{pm}$$

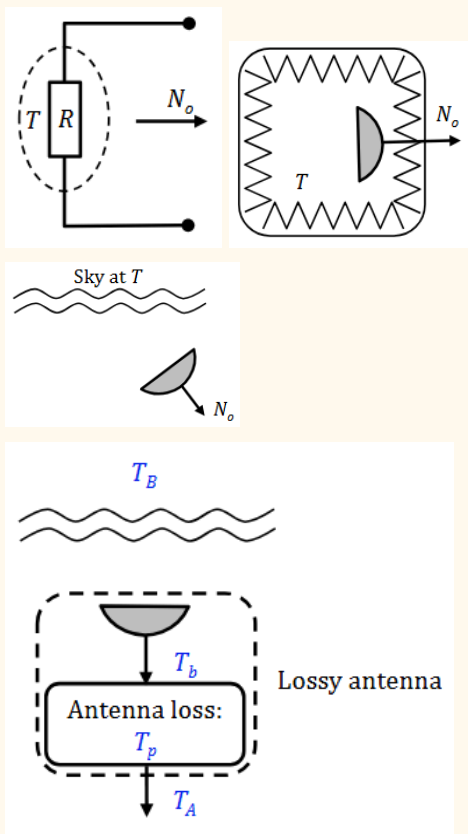
$$U_{incident} = G_{tx} \frac{P_{tx}}{4\pi r^2}$$

$$A_{e, rx} = G_{rx} \frac{\lambda^2}{4\pi}$$

$$P_{rx} = P_{tx} G_{tx} G_{rx} \left(\frac{\lambda}{4\pi r}\right)^2 e_{pm}$$

$$P_{rx, dBW} = P_{tx, dBW} + G_{tx, dB} + G_{rx, dB} + 20 \log\left(\frac{\lambda}{4\pi r}\right) + 20 \log(|\hat{a}_{incident} \cdot \hat{a}_{rx}|)$$

WARNING: 4π not 2π !!!



$$N_o = kTB$$

- no matter resistor, antenna in anechoic chamber, or antenna in anywhere, the thermal noise power generated to a match load is the same, if the physical temperature is the same
- The noise generated is the same, but the equivalent temperature is different, since it depends on e_{rad}
 - e_{rad} does not come into consideration to thermal noise power generation, because thermal noise depends solely, on physical temperature and bandwidth (like resistor thermal noise power does not depends on R)
 - BUT e_{rad} does come into consideration for antenna equivalent noise temperature, T_A

$T_B(\theta, \phi) = \text{background noise temperature}$

ground = 290–300 K, zenith = 3–5 K, horizon = 50–100 K

- the equivalent noise temperature received from each direction

$T_b = \text{brightness temperature}$

$$T_b = \frac{\int_{\theta=0, \phi=0}^{\pi} \int_0^{2\pi} T_B(\theta, \phi) D(\theta, \phi) \sin\theta \, d\phi \, d\theta}{\int_{\theta=0, \phi=0}^{\pi} \int_0^{2\pi} D(\theta, \phi) \sin\theta \, d\phi \, d\theta}$$

- average of the background noise temperature taking antenna directivity into account

$$T_A = e_{rad} T_b + (1 - e_{rad}) T_p$$

$e_{rad} T_b$ from background

$(1 - e_{rad}) T_p$ from thermal noise

- when antenna is 100% efficient, all antenna noise temperature T_A is from background
- when antenna is 0% efficient, all antenna noise temperature T_A is from thermal noise like resistor

Gain Over Temperature

Figure of merit given to the receiving antenna.

Can also be for the entire Rx chain, including all the gain and equivalent temperature, T_e

$$G/T_{dB} = 10 \log(G_{rx, linear} / T_A) = G_{rx, dBi} - 10 \log(T_{A \text{ or } E}) \quad \text{in dB/K}$$

Signal power delivered by a Rx antenna to a matched receiver input (somehow, no need to divide by 2)

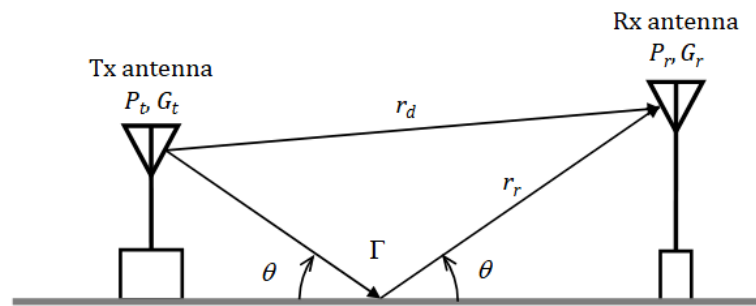
$$S_{input} = P_{rx} = P_{tx} G_{tx} G_{rx} \left(\frac{\lambda}{4\pi r}\right)^2 e_{pm}$$

SNR input to receiver

$$SNR = \frac{S_{input}}{N_{input}} = \frac{P_{tx} G_{tx} G_{rx} \left(\frac{\lambda}{4\pi r}\right)^2 e_{pm}}{k T_A B} = \frac{G_{rx}}{T_A} \frac{P_{tx} G_{tx} \left(\frac{\lambda}{4\pi r}\right)^2 e_{pm}}{kB}$$

Receiving antenna G/T is the only controllable factor in receiving antenna to optimise for performance

Reflection



Geometry of a radio link

• Direct path:

- Friis equation for pathloss decreasing as $1/r^2$ in free space
- Received voltage at receiver

$$V_d = \sqrt{P_r Z_l} e^{-jk_o r_d} = \frac{\sqrt{G_r G_t P_t Z_l} \lambda}{4\pi r_d} e^{-jk_o r_d}$$

• Reflection path

- Friis equation but wave is specularly reflected by ground with coefficient Γ (equal angles θ of incidence and reflection by Snell's Law)
- Received voltage at receiver: $V_r = \Gamma \sqrt{P_r Z_l} e^{-jk_o r_r} = \Gamma \frac{\sqrt{G_r G_t P_t Z_l} \lambda}{4\pi r_r} e^{-jk_o r_r}$

• Total received voltage at receiver if the gain at direct and reflection directions are equal and $r_d \approx r_r \approx d$

$$V = V_r + V_d = \frac{\sqrt{G_r G_t P_t Z_l} \lambda}{4\pi} \left(\Gamma \frac{e^{-jk_o r_r}}{r_r} + \frac{e^{-jk_o r_d}}{r_d} \right)$$

$$\cong \frac{\sqrt{G_r G_t P_t Z_l} \lambda}{4\pi} \frac{e^{-jk_o r_d}}{r_d} [\Gamma e^{-jk_o(r_r - r_d)} + 1]$$

$F = |\Gamma e^{-jk_o(r_r - r_d)} + 1|$: path gain factor

- affected by locations of transmitter and receiver as well as ground (Γ).
- $0 \leq F \leq 2$: The power received by receiver may be cancelled or enhanced to four times.

An transmit antenna of a radio system operating at 1800MHz is installed at a height of 8.33 m. The distance between the transmitter and receiver is 1000 m. Find (1) the smallest receive antenna height to maximize the receive signal voltage and (2) received power if the reflection coefficient of ground is $\Gamma = -1$.

$\lambda = 0.167$ m at 1800 MHz. $d = 1000$ m. As $d \gg (h_1, h_2)$ with Taylor expansion of $\sqrt{1+x} \cong 1 + \frac{x}{2}$

$$r_r = \sqrt{d^2 + (h_1 + h_2)^2} \cong d + \frac{(h_1 + h_2)^2}{2d} \text{ and } r_d = \sqrt{d^2 + (h_2 - h_1)^2} \cong d + \frac{(h_2 - h_1)^2}{2d}$$

Maximize path gain factor F as $\Gamma = -1$ $r_d \approx r_r \approx d$, $F = |\Gamma e^{-jk_o(r_r - r_d)} + 1| = 2 \left| \sin \left(\frac{k_o h_1 h_2}{d} \right) \right|$

$$\frac{k_o h_1 h_2}{d} = \frac{\pi}{2} + n\pi \quad n=0, 1, 2, \dots \text{ then } h_2 = \frac{\pi}{2} \frac{d}{k_o h_1} = \frac{\pi}{2} \frac{1000 (0.167)}{2\pi (8.33)} = 5 \text{ m}$$

$$\text{Received power: } P_r = \frac{|V|^2}{Z_l} \cong \frac{G_t P_t G_r h_1^2 h_2^2}{d^2 r_d^2} \cong \frac{G_t P_t G_r h_1^2 h_2^2}{r_d^4}$$

As $2 \left| \sin \left(\frac{k_o h_1 h_2}{d} \right) \right| \cong 2 \left(\frac{k_o h_1 h_2}{d} \right)$ with Taylor series approximation $\sin x \cong x$ and $r_d \approx d$.

The pathloss is r^4 for this ideal case.

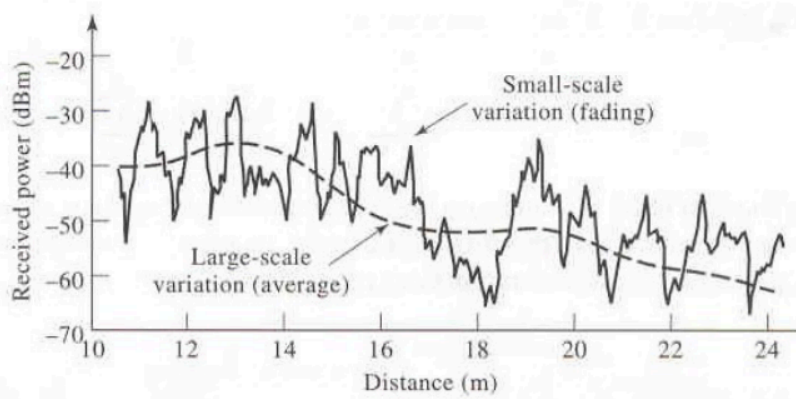
So, pathloss is r^n , $n=2$ for free space, $n>4$ for more complicated cases.

Fading

Fading

- Multipath signals arrive at receiver with different amplitudes and phases, the net received signal voltage vary due to destructive interference.
- Long path lengths, rapid time variations in received amplitude occur over distances as short as $\lambda/2$.
- User, scatterers/reflectors moving, these variation occur over relatively short time intervals.

Beside large amplitude variations, fading can also involve frequency modulation due to Doppler effects, and variable signal delays caused by time-varying propagation paths.



small-scale effect (fading):

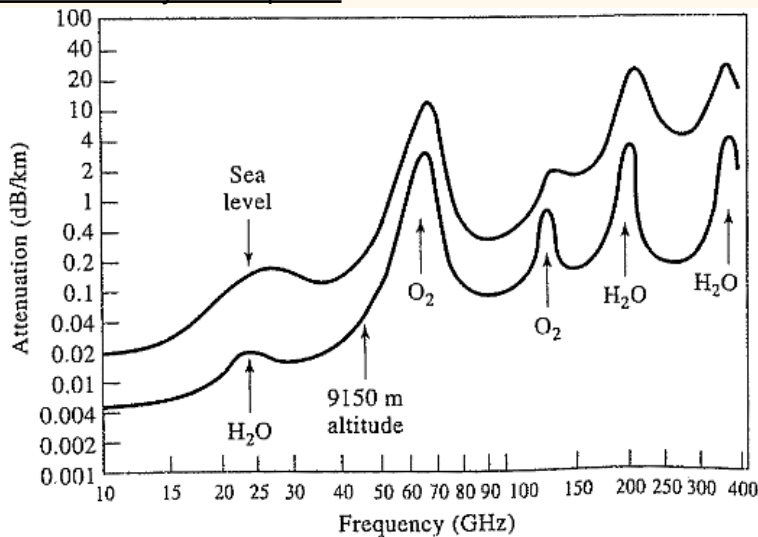
involving large variations in amplitude over small distances or time intervals.

large-scale effects:

caused by $1/R^n$ path losses, or blockage or other effects that produce slower variations over relatively large distances.

To improve reliability against fading use noncorrelative channels: spatial, polarisation and pattern diversity

Attenuation by Atmosphere



Average atmospheric attenuation versus frequency (horizontal polarization).

- If P_{in} is 1W = 30dBm and $RL = 10$ dB
 - $S_{11} = -10$, $S_{11} = \sqrt{0.1}$, 10% power is reflected back
 - Preflected is 10dB down from 30dBm = 20dBm = 0.1W
 - Power transferred 0.9W = 29.5dBm
 - $|\Gamma| = S_{11} = \sqrt{0.1} \Rightarrow VSWR = 1.92:1$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$|\Gamma| = \frac{VSWR - 1}{VSWR + 1}$$

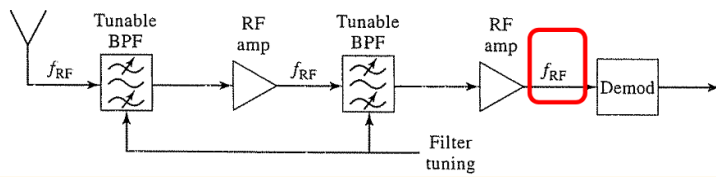
$$P_{reflected} = |\Gamma|^2 P_{in}$$

$$RL_{dB} = 10 \log\left(\frac{P_{in}}{P_{reflected}}\right) = -20 \log(|\Gamma|)$$

$$|s_{11}|_{dB} = 20 \log(|s_{11}|) = 20 \log(|\Gamma|) = -RL_{dB}$$

Receiver Chain

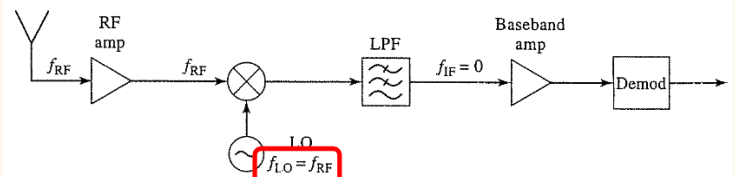
Basic Architecture



(a) Tuned radio frequency (TRF)

- One of the earliest types of receiving circuits
- Stages of RF amplifiers + tunable bandpass filters for high gain and selectivity or combined filter and amplifier.
- Poor selection and limited gain
- Broadcasting in early days

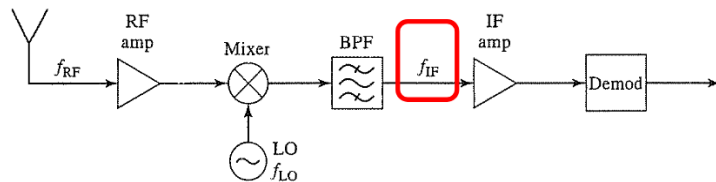
- good for old times with very low frequency
- the stacking filter and amplifier is to try to get rid all the out of band noise
- there is no down conversion, just raw RF



(b) Direct conversion receiver

- Mixer and local oscillator for frequency down-conversion with a zero IF frequency (baseband). LO frequency=RF.
- Simpler and less costly than super heterodyne receivers, no IF amplifier, IF bandpass filter, or IF local oscillator required, no image frequency
- Disadvantage: LO with a very high degree of precision and stability,
- Doppler radars, exact LO obtained from transmitter

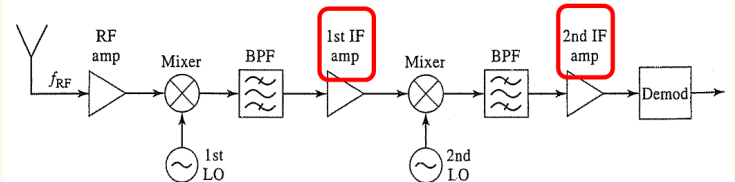
- there is no IF
- not good performance but use for analog data
- the no image frequency is only true if the sidebands around the carrier is symmetrical
 - unless zero second IF is used with quadrature down conversion



(c) Super-heterodyne Receiver

- Most popular type of receiver for 50 years
- Similar to (b) but nonzero IF frequency
- With midrange IF, sharper cutoff filters for improved selectivity, and higher IF gain through the use of an IF amplifier.
- Majority of broadcast radios and televisions, radar systems, cellular telephone systems, and data communications systems.

- IF components are cheaper so cost not much an issue
 - at IF (a lower frequency) narrowband filtering and amplifying is easier
- this is the tuning LC filter to tune to carrier (TRF) VS having a very good narrow band filter and use the mixer to down convert to that location
- trade off with high vs low IF, at high IF, image rejection easier but channel selection harder

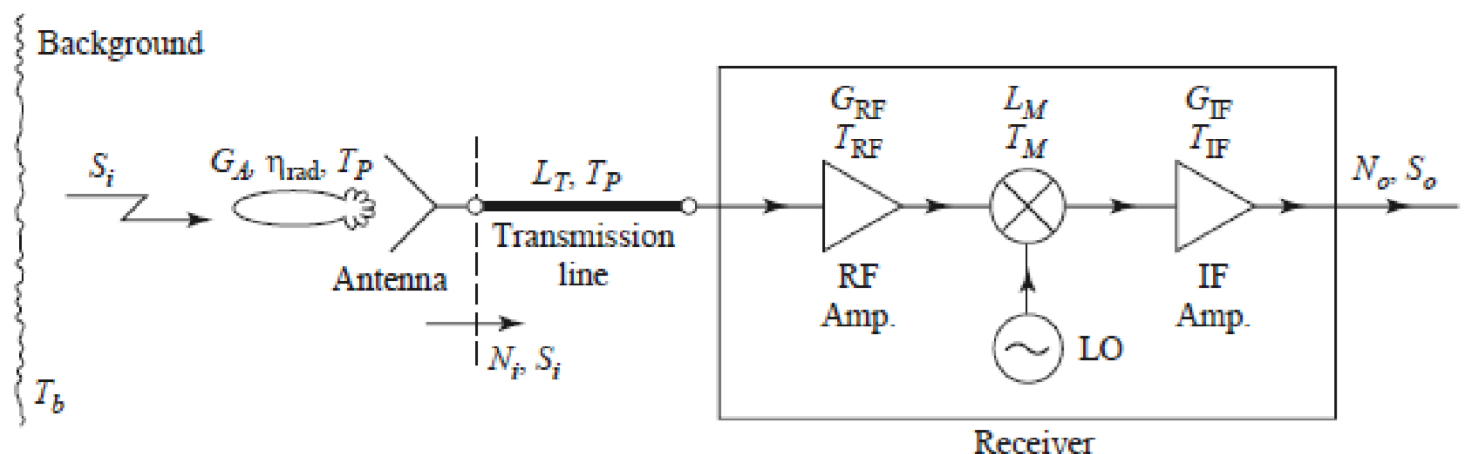


(d) Dual-conversion superheterodyne receiver

- At microwave and millimeter wave frequencies
- Two stages of down conversion to avoid problems due to LO stability.

- By doing the downconversion step by step,
 - IF1 can be large $\Rightarrow$ easy to filter off image in RF, since image is far away
 - IF2 can be small $\Rightarrow$ easy to do narrow band channel selection filter without unreasonable high Q filter

Receiver Noise



(a) Noise characteristics of a complete antenna–transmission line– receiver front end

Total noise power at the output of the receiver, N_o , from

- Antenna pattern and loss
- Loss in the transmission line
- Receiver components

To determine minimum detectable signal level for receiver and maximum link range for a given transmitter power

- Receiver components consist of
 - an RF amplifier with gain G_{RF} and noise temperature T_{RF}
 - a mixer with an RF-to-IF conversion loss factor L_M and noise temperature T_M
 - an IF amplifier with gain G_{IF} and noise temperature T_{IF}
- Noise effects of later stages usually ignored
- Overall noise figure dominated by the characteristics of the first few stages.
- Component noise temperatures related to noise figures as $T = (F-1)T_0$.
- Equivalent noise temperature of the receiver can be found as: $T_{REC} = T_{RF} + T_M/G_{RF} + L_M T_{IF}/G_{RF}$
- Equivalent noise temperature of transmission line connecting antenna to receiver has a loss L_T at a physical temperature T_p : $T_{TL} = (L_T - 1)T_p$
- Noise temperature of the transmission line (TL) and receiver (REC) cascade is

$$T_{TL+REC} = T_{TL} + L_T T_{REC} = (L_T - 1)T_p + L_T T_{REC}$$

- mixer is lossy
- loss is inverse of gain so in cascading its reciprocal

- Noise temperature of the antenna (mainbeam only):

$T_A = e_{rad}T_b + (1 - e_{rad})T_p$
 e_{rad} is the efficiency of the antenna, T_p is its physical temperature, and T_b is the equivalent brightness temperature of the background seen by the mainbeam. The noise power at the antenna terminals, which is also the noise power delivered to the transmission line, is

$$N_i = kBT_A = kB[e_{rad}T_b + (1 - e_{rad})T_p],$$

where B is the system bandwidth.

- If S_i is the received power at the antenna terminals, then the input SNR at the antenna terminals is S_i/N_i . The output signal power is

$$S_o = S_i G_{RF} G_{IF} / (L_T L_M) = S_i G_{SYS}$$

where G_{SYS} is a system power gain.

- The output noise power is

$$N_o = (N_i + kBT_{TL+REC})G_{SYS} = kB(T_A + T_{TL+REC})G_{SYS} = kB[e_{rad}T_b + (1 - e_{rad})T_p + (L_T - 1)T_p + L_T T_{REC}]G_{SYS} = kBT_{SYS}G_{SYS}$$

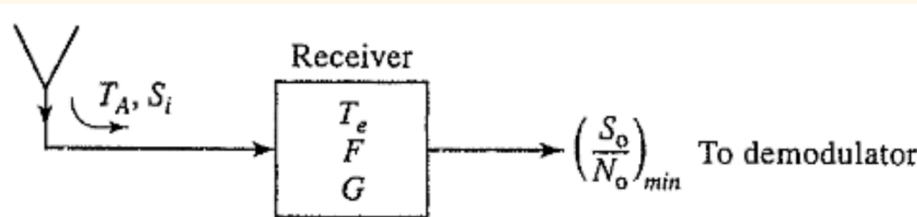
where T_{SYS} has been defined as the overall system noise temperature.

- The output SNR is

$$S_o/N_o = S_i / (kBT_{SYS}) = S_i / \{kB[e_{rad}T_b + (1 - e_{rad})T_p + (L_T - 1)T_p + L_T T_{REC}]\}$$

G_{sys} doesn't improve SNR

- this all is to find the output SNR
 - used for the minimum SNR required
 - then can find the minimum Tx power required
- receiver can never improve the SNR, can only avoid additional noise



- Output signal power and total output noise power:

$$S_o = G S_i \text{ and } N_o = kBG(T_A + T_e) \text{ where } B: \text{bandwidth of receiver (set by IF bandpass filter)}$$

- Minimum detectable input signal level:

$$S_{i_{min}} = \frac{S_{o_{min}}}{G} = \left(\frac{N_o}{G}\right) \left(\frac{S_o}{N_o}\right)_{min} = kB(T_A + T_e) \left(\frac{S_o}{N_o}\right)_{min}$$

- From the minimum SNR required by the demodulator, and known noise figure of the front end, the minimum input signal power can be found, this is called Minimum Detectable Signal, MDS
- This is the minimum signal power the antenna must give to the front end
- Since the basis is from SNR, increasing the gain of the receiver front end does not help with MDS (but a minimum signal power must exist to the demodulator, so the gain is needed to give S_o in the mW range)
- Increasing Rx antenna gain can help
- Using MDS and radio link equation (Friis or ground reflection) can find the minimum Tx power

• *Receiver voltage sensitivity (in short sensitivity):*

Minimum detectable signal voltage (time-average) converted from minimum detectable signal power for a given receiver input impedance of Z_o , e.g. $50\ \Omega$:

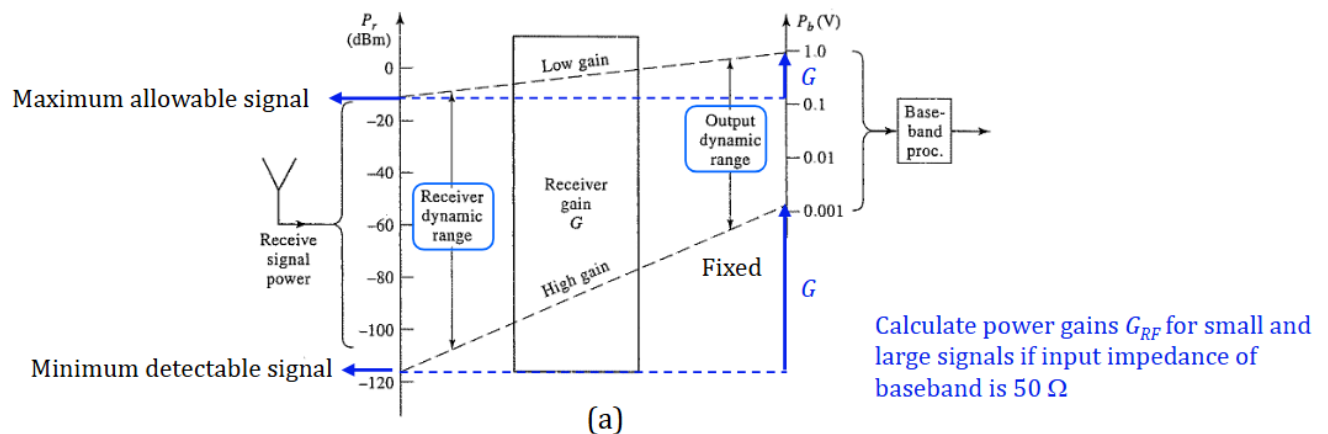
$$V_{i_{min}} = \sqrt{2Z_o S_{i_{min}}} \text{ V}$$

• *Receiver dynamic range, DR_r :*

The range from the minimum detectable signal level to maximum allowable signal level

$$DR_r = \frac{\text{maximum allowable signal power}}{\text{minimum detectable signal power}}$$

- sensitive or receiver voltage sensitivity, is just the MDS converted to voltage using Z_0 , noting the 2 for rms, so the voltage is V_{peak} or $V_{amplitude}$, 10 mW at $50\ \Omega$ is 1 V_{peak}
- the maximum allowable input signal power is limited by the saturation along the chain



Automatic gain control (AGC) : a circuit with variable-gain function for a wide range of input signal levels. AGC often implemented **at IF stage or RF stage**.

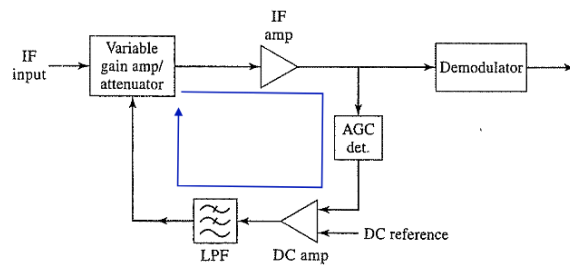
- Receiver gain: 80-100 dB for minimum detectable signal of 10 mW (about 1 V peak at $50\ \Omega$)
- Gain distributed throughout the RF, IF, and baseband stages **to avoid oscillation**
- Most of gain occurring at IF stage
 - Amplifiers and other components cheaper at lower frequencies.
 - Avoid very high levels of gain at RF (front-end) (high input signal levels > 1-dB compression point (P_1), or 3rd-order intercept point (P_3) of front-end components)
 - A moderate level of gain at RF stage to set a good noise figure for receiver (but)
- Dynamic range at output of receiver: 60 dB smaller than 80-100 dB dynamic range at receiver input.

(a) Input and output dynamic ranges of a typical receiver.

Power gain through receiver varies as a function of input signal strength to fit signal range into baseband processing range

- A receiver gain on the order of 60 dB may be required for low level signals
- A gain of only 20-30 dB may be required for high-level signals.

- AGC is used for increasing the receiver dynamic range, by mapping the bigger range to the smaller range of demodulator dynamic range



(a)

(a) IF AGC circuit

- A variable voltage-controlled attenuator (or variable gain amplifier)
- A detector to convert a sample of IF voltage to a DC value
- Rectified signal is compared with a DC reference level
- Output of low-pass filter controls the variable attenuator
- Increase in control voltage increases attenuation.
- Dynamic range of the AGC sets the required range of attenuation, typically 20~30 dB
- Loss (due to attenuator) increase thermal noise of IF stage, but signal high
- If properly designed, and spread over two or more stages of amplification, an AGC circuit can provide improved SNR as the input signal level increases.

